

Bayesian Expression Recovery in Bulk RNA-Seq: Assessing Prior Choices Under a Poisson–Gamma Framework

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Team 6 —June 2025

Abstract

Dropouts and amplification noise obscure true transcript abundances in RNA-sequencing, motivating statistical “expression recovery” before downstream analysis. While single-cell methods such as SCIMPUTE [1], SAVER [2], and its transfer-learning extension SAVER-X [3] have been proved effective, their assumptions and performance in *bulk* data remain under-explored.

We formulate a unified Poisson–Gamma hierarchical model and embed four alternative priors—(i) non-informative, (ii) empirical two-component mixture (EM), (iii) Poisson-Lasso (SAVER), and (iv) autoencoder-informed (SAVER-X)—to recover latent gene expression. The pipeline is evaluated on a 140-sample pancreatic ductal adenocarcinoma (PDAC) bulk RNA-seq cohort and on synthetic drop-out series ($p = 0.05\text{--}0.8$). Recovery accuracy is quantified by R^2 , Pearson correlation, and RMSE against ground truth, with uncertainty assessed via posterior predictive checks.

In benchmark experiments spanning 5–80 % synthetic dropout, the auto-encoder-informed prior (SAVER-X) achieved the highest R^2 ($0.96 \rightarrow 0.12$) and the lowest RMSE ($0.20 \rightarrow 0.94$), retaining clear superiority even under severe sparsity. The non-informative prior remained competitive at light dropout yet deteriorated rapidly as zeros accumulated, while Poisson-Lasso (SAVER) suffered from outlier sensitivity and the EM two-component mixture consistently under-imputed counts. These patterns indicate that informative external priors and adaptive variance modelling are critical for reliable bulk RNA-seq recovery.

Collectively, our unified Poisson–Gamma analysis positions SAVER-X as the current best-practice Bayesian solution for bulk RNA-seq imputation, while underscoring the need for fair benchmarking with matched external data and the exploration of richer hierarchical priors to generalise these gains.

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1 Introduction

1.1 Motivation

RNA sequencing (RNA-Seq) technologies have revolutionised transcriptomic profiling by providing high-resolution gene expression data. However, due to experimental limitations, widespread dropouts occur in both bulk (tissue or organism resolution) and single-cell RNA-Seq data. In practice, zero or near-zero observed counts occur in many genes whose original expression values are non-zero. This may happen at a frequency of more than 90%, leading to substantial challenges in downstream analysis.

Robust recovery (*imputation*) of true expression values is therefore crucial for downstream analyses. In single-cell RNA-Seq, thanks to homogeneity across cells, identifiable cell types, and batch effects, various imputation methods have been developed. However, in *bulk* RNA-Seq, the lack of such structure poses unique challenges for imputation. Specifically, single-cell imputation methods often rely on cell-specific priors, which are not directly applicable to bulk data.

As for our project, we focus on the Bayesian framework for bulk RNA-Seq imputation, which has been widely adopted in single-cell RNA-Seq. We aim to systematically evaluate different Bayesian-based pre-existing methods within a unified Poisson–Gamma hierarchical model. Moreover, by selecting different priors on a unified framework, we can compare their performance in terms of recovery accuracy and computational efficiency, hoping to provide insights into the best practices for bulk RNA-Seq imputation.

1.2 Objectives

This study systematically evaluates four Bayesian priors within a unified Poisson–Gamma hierarchical model for bulk RNA-Seq noise correction and expression recovery. Specifically, we compare:

1. **Non-informative prior** (empirical Bayes and small- ε alternatives), such priors are often used in practice to avoid overfitting;
2. **Two-component mixture prior** estimated via an Expectation–Maximisation (EM) algorithm, which is adopted in the scImpute method;
3. **Poisson–Lasso prior** underlying the original SAVER method, which imposes sparsity on the latent expression values;
4. **Autoencoder prior** as employed in SAVER-X, which uses pretrained data to inform the imputation process.

Using real bulk RNA-Seq data, we examine their recovery accuracy across a range of simulated dropout rates using quantitative metrics and visual diagnostics.

1.3 Contribution

Our work revisits prior selection in a bulk context, highlights strengths and shortcomings of each approach, and outlines best-practice methods in our experiments. By performing quantitative and visual evaluations, we explain their relative performance and analyze potential reasons for their behavior.

2 Materials and Methods

2.1 Data Acquisition and Pre-processing

- **Dataset:** 140 pancreatic ductal adenocarcinoma (PDAC) tissues from Cao et al. [4]. Raw counts underwent $\log_2(x+1)$ transformation prior to public release.
- **Filtering:** Genes expressed in all tissues were retained (18,022 genes). i.e., we discarded all genes that fail to express (the data becomes 0) in at least one patient.
- **Normalisation:** Z-score per gene across samples.
- **Dropout Simulation:** Independent masking probabilities $p \in \{0, 0.05, 0.10, 0.20, 0.40, 0.60, 0.80\}$ to generate missing-at-random patterns.

Ground truth is defined as the unmasked matrix; masked observations constitute the test input.

2.2 Notation and Likelihood

Let Y_{cg} denote the observed (possibly masked) read count for gene g ($g = 1, \dots, G$) in patient (sample) c ($c = 1, \dots, C$). After gene-wise $\log_2(x+1)$ transformation and *patient-wise* z -scaling, library-size effects are largely removed, so we adopt the simplified Poisson data model

$$Y_{cg} \mid \lambda_{cg} \sim \text{Poisson}(\lambda_{cg}), \quad \lambda_{cg} > 0, \quad (1)$$

where the latent intensity λ_{cg} is the *true* (unobserved) expression level. All four recovery methods differ *only* in the prior distribution placed on λ_{cg} .

Following the SAVER framework of Huang et al. [2], the posterior mean $\hat{\lambda}_{cg} = \mathbb{E}[\lambda_{cg} \mid Y_{cg}]$ serves as the point estimate supplied to downstream analyses. In what follows, we denote by $\hat{\lambda}_{cg}^{(m)}$ the estimator under model $m \in \{1, \dots, 4\}$.

2.3 Prior Families

2.3.1 Non-informative Prior (PG-NI)

For each gene g , we retain the conjugate Poisson–Gamma hierarchy under the *shape–rate* parameterisation,

$$Y_{cg} \mid \lambda_{cg} \sim \text{Poisson}(\lambda_{cg}), \quad \lambda_{cg} \sim \text{Gamma}(\alpha_g, \beta_g),$$

with $\alpha_g > 0$ (shape) and $\beta_g > 0$ (rate). In order to inject minimal prior information while preserving conjugacy, we investigate two weak-prior settings:

- **Small- ε prior.** We set $\alpha_g = \beta_g = \varepsilon$ with a small constant $\varepsilon = 10^{-3}$. This yields a proper prior whose density is nearly flat on the log-scale and numerically approximates the Jeffreys prior $\pi(\lambda) \propto \lambda^{-1/2}$.
- **Empirical-Bayes (EB) moment matching.** Denote the sample mean and variance of gene g across the C patients by $\bar{Y}_g = \frac{1}{C} \sum_{c=1}^C Y_{cg}$ and $s_g^2 = \frac{1}{C-1} \sum_{c=1}^C (Y_{cg} - \bar{Y}_g)^2$. Matching $\mathbb{E}[\lambda_{cg}] = \bar{Y}_g$ and $\text{Var}[\lambda_{cg}] = s_g^2 - \bar{Y}_g$ gives

$$\alpha_g^{\text{EB}} = \frac{\bar{Y}_g^2}{s_g^2 - \bar{Y}_g}, \quad \beta_g^{\text{EB}} = \frac{\bar{Y}_g}{s_g^2 - \bar{Y}_g}, \quad (2)$$

provided $s_g^2 > \bar{Y}_g$. When this condition fails (rare for log-transformed counts), we revert to the Small- ε prior in (a).

Posterior distribution. Under either choice of (α_g, β_g) , conjugacy yields

$$\lambda_{cg} \mid Y_{cg} \sim \text{Gamma}(\alpha_g + Y_{cg}, \beta_g + 1). \quad (3)$$

Point estimator. Throughout this study, we adopt the posterior mean

$$\hat{\lambda}_{cg}^{(\text{NI})} = \mathbb{E}[\lambda_{cg} \mid Y_{cg}] = \frac{\alpha_g + Y_{cg}}{\beta_g + 1}. \quad (4)$$

Note that as $\alpha_g, \beta_g \rightarrow 0$ the estimator reduces to Y_{cg} , while for small counts it provides adaptive shrinkage toward the gene-level prior mean.

2.3.2 Two-Component Poisson-Gamma Mixture (PG-Mix)

Model definition. For each gene g and patient c we introduce an unobserved *drop-out indicator* $Z_{cg} \in \{0, 1\}$, with $Z_{cg} = 0$ interpreted as a genuine drop-out (very low expression) and $Z_{cg} = 1$ as an expressed state. Conditional on the indicator, we place a Gamma prior on the latent mean λ_{cg} , while the observed count Y_{cg} follows a Poisson likelihood:

$$Z_{cg} \sim \text{Bernoulli}(\pi_g), \quad (M1)$$

$$\lambda_{cg} \mid Z_{cg} = k \sim \text{Gamma}(\alpha_g^{(k)}, \beta_g^{(k)}), \quad k \in \{0, 1\}, \quad (M2)$$

$$Y_{cg} \mid \lambda_{cg} \sim \text{Poisson}(\lambda_{cg}). \quad (M3)$$

Hence the marginal distribution of Y_{cg} is the **two-component negative binomial mixture**

$$p(Y_{cg} = y) = \pi_g \text{NB}(y; \alpha_g^{(0)}, \beta_g^{(0)}) + (1 - \pi_g) \text{NB}(y; \alpha_g^{(1)}, \beta_g^{(1)}),$$

where, under the shape-rate convention,

$$\text{NB}(y; \alpha, \beta) = \binom{y + \alpha - 1}{y} \left(\frac{\beta}{1 + \beta} \right)^\alpha \left(\frac{1}{1 + \beta} \right)^y, \quad y \in \mathbb{N}.$$

EM algorithm. We estimate $(\pi_g, \alpha_g^{(k)}, \beta_g^{(k)})$ *gene-by-gene* via Expectation-Maximisation, initialising (α, β) by method-of-moments on $\{Y_{cg}\}_{c=1}^C$ and $\pi_g = 0.5$.

E-step. Compute the posterior responsibility (the probability that sample c belongs to the *drop-out* component):

$$w_{cg} = \Pr(Z_{cg} = 0 \mid Y_{cg} = y) = \frac{\pi_g \text{NB}(y; \alpha_g^{(0)}, \beta_g^{(0)})}{\pi_g \text{NB}(y; \alpha_g^{(0)}, \beta_g^{(0)}) + (1 - \pi_g) \text{NB}(y; \alpha_g^{(1)}, \beta_g^{(1)})}. \quad (E1)$$

M-step. Update parameters by weighted method-of-moments (WMoM):

$$\pi_g^{\text{new}} = \frac{1}{C} \sum_{c=1}^C w_{cg}, \quad (5)$$

$$(\alpha_g^{(0)}, \beta_g^{(0)}) \leftarrow \text{WMoM}(\{w_{cg}, Y_{cg}\}_{c=1}^C), \quad (6)$$

$$(\alpha_g^{(1)}, \beta_g^{(1)}) \leftarrow \text{WMoM}(\{1 - w_{cg}, Y_{cg}\}_{c=1}^C). \quad (M1-M3)$$

Concretely, given weights $\{u_c\}$ and data $\{y_c\}$, WMoM solves $\hat{\mu} = \sum u_c y_c / \sum u_c$, $\hat{\sigma}^2 = \sum u_c (y_c - \hat{\mu})^2 / \sum u_c$ and plugs into $\alpha = \hat{\mu}^2 / (\hat{\sigma}^2 - \hat{\mu})$, $\beta = \hat{\mu} / (\hat{\sigma}^2 - \hat{\mu})$ (same formulas as (2)).

Posterior distribution and estimator. Conditional on the final parameters, the posterior of λ_{cg} is a *mixture of two Gamma posteriors*:

$$\lambda_{cg} \mid Y_{cg} = y \sim w_{cg} \Gamma(\alpha_g^{(0)} + y, \beta_g^{(0)} + 1) + (1 - w_{cg}) \Gamma(\alpha_g^{(1)} + y, \beta_g^{(1)} + 1), \quad (7)$$

$$\hat{\lambda}_{cg}^{(\text{Mix})} = w_{cg} \frac{\alpha_g^{(0)} + y}{\beta_g^{(0)} + 1} + (1 - w_{cg}) \frac{\alpha_g^{(1)} + y}{\beta_g^{(1)} + 1}. \quad (\text{Est})$$

This estimator smoothly interpolates between an “aggressive shrinkage” regime (drop-out component) and a “mild shrinkage” regime (expression component), adapting to per-sample evidence.

Implementation tips.

- Use `logsumexp` to evaluate responsibilities in (E1) for numerical stability when y is large.
- Constrain $\alpha_g^{(k)}, \beta_g^{(k)} \geq 10^{-3}$ to avoid degenerate variance estimates during early EM iterations.
- Pre-compute $\log \Gamma(\alpha)$ tables for common shape values to speed up the negative binomial pmf.

2.3.3 Poisson–Lasso Prior (SAVER)

Likelihood. After patient-wise z -scaling we set the library size normaliser $s_c = 1$, so the observation model is

$$Y_{cg} \mid \lambda_{cg} \sim \text{Poisson}(\lambda_{cg}). \quad (\text{L1})$$

Step 1: Predictive mean via Poisson–Lasso. For each gene g we form the predictor vector $\mathbf{X}_{c,-g} = (\log(Y_{cg'} + 1))_{g' \neq g}$ and solve the *penalised Poisson regression*

$$\min_{\boldsymbol{\beta}_g} \left\{ - \sum_{c=1}^C [Y_{cg} (\mathbf{X}_{c,-g}^\top \boldsymbol{\beta}_g) - \exp(\mathbf{X}_{c,-g}^\top \boldsymbol{\beta}_g)] + \lambda \|\boldsymbol{\beta}_g\|_1 \right\}, \quad (\text{P1})$$

where λ is selected by 5-fold **LassoCV** on a subset of representative genes (as in the original SAVER implementation). The fitted linear predictor gives the *prior mean*

$$\mu_{cg} = \exp(\mathbf{X}_{c,-g}^\top \hat{\boldsymbol{\beta}}_g). \quad (\text{P2})$$

Step 2: Gene-specific dispersion (variance) factor. Following Huang et al. [2], we estimate the over-dispersion parameter $\phi_g > 0$ by method-of-moments on $\{Y_{cg}\}_{c=1}^C$:

$$\phi_g = \max\left\{0, \frac{1}{\bar{Y}_g^2} (s_g^2 - \bar{Y}_g)\right\}, \quad \bar{Y}_g = \frac{1}{C} \sum_c Y_{cg}, \quad s_g^2 = \frac{1}{C-1} \sum_c (Y_{cg} - \bar{Y}_g)^2. \quad (\text{D1})$$

Empirically truncating at 0 avoids negative estimates when Poisson variance dominates.

Step 3: Gamma prior parameters. The moment equations $\mathbb{E}[\lambda_{cg}] = \mu_{cg}$ and $\text{Var}[\lambda_{cg}] = \phi_g \mu_{cg}^2$ are matched by the Gamma distribution $\lambda_{cg} \sim \Gamma(\alpha_{cg}, \beta_{cg})$ with

$$\alpha_{cg} = \phi_g^{-1}, \quad \beta_{cg} = \frac{\alpha_{cg}}{\mu_{cg}} = \frac{1}{\phi_g \mu_{cg}}. \quad (\text{G1})$$

Posterior distribution. Conjugacy with likelihood (L1) yields

$$\lambda_{cg} \mid Y_{cg} = y \sim \Gamma(\alpha_{cg} + y, \beta_{cg} + 1). \quad (\text{G2})$$

Posterior mean (shrinkage form). The posterior mean can be expressed as a convex combination of the *observed count* and the *predicted prior mean*:

$$\hat{\lambda}_{cg}^{(\text{SAV})} = \frac{\alpha_{cg} + Y_{cg}}{\beta_{cg} + 1} \quad (8)$$

$$= \underbrace{\frac{1}{\beta_{cg} + 1}}_{w_{\text{obs}}} Y_{cg} + \underbrace{\frac{\beta_{cg}}{\beta_{cg} + 1}}_{w_{\text{pred}}} \mu_{cg}, \quad (\text{PM})$$

where

$$w_{\text{obs}} = \frac{1}{1 + \beta_{cg}}, \quad w_{\text{pred}} = 1 - w_{\text{obs}} = \frac{\beta_{cg}}{1 + \beta_{cg}}. \quad (\text{SW})$$

Because $\beta_{cg} = 1/(\phi_g \mu_{cg})$, genes with low dispersion ($\phi_g \ll 1$) or high predicted mean receive *less* shrinkage ($w_{\text{obs}} \approx 1$); conversely, sparse genes with noisy predictions are heavily pulled toward μ_{cg} .

2.3.4 Autoencoder Prior (SAVER-X)

Rationale. SAVER-X [3] leverages a *denoising autoencoder* (AE) to learn cross-gene co-expression patterns from an external reference cohort and uses the reconstructed values as data-driven priors. In our context, non-PDAC pancreatic samples serve as an unbiased source of “healthy” expression, while the target cohort comprises PDAC tumours with synthetic drop-out (Section 2.1).

Network architecture. Let $n = 18,022$ be the number of intersecting genes. We employ a symmetrical fully-connected AE:

$$\underbrace{\text{Input}(n)} \xrightarrow{1024 \text{ ReLU}} \xrightarrow{\text{Dropout } 0.3} 256 \text{ ReLU} \xrightarrow{\text{Dropout } 0.3} \xrightarrow{\text{Latent}(32)} 256 \text{ ReLU} \xrightarrow{\text{Dropout } 0.3} \xrightarrow{1024 \text{ ReLU}} \xrightarrow{\text{Linear}(n)} \text{Reconstruction}$$

with Adam optimiser ($\eta = 10^{-3}$), mean-squared-error loss and early stopping (**patience**=20). All layers use He initialisation.

Transfer-learning protocol.

1. **Pre-training** on $C_{\text{pre}} = 212$ non-PDAC samples (healthy and pancreatitis tissues) for up to 200 epochs, mini-batch size 256.
2. **Fine-tuning** on $C_{\text{tar}} = 140$ PDAC tumour samples (containing 5–80 % simulated drop-out) for up to 50 epochs, mini-batch size 16.

Prior mean. Let $\tilde{X}_{cg}$ denote the AE reconstruction of Y_{cg} . We set the prior $\lambda_{cg} \mid \tilde{X}_{cg} \sim \Gamma(\alpha_{cg}, \beta_{cg})$ with $\alpha_{cg} = 1/\phi_g$, $\beta_{cg} = 1/(\phi_g \tilde{X}_{cg})$, where ϕ_g is the gene-wise dispersion from Eq. (D1). Thus the prior *mean* equals the AE output: $\mathbb{E}[\lambda_{cg}] = \tilde{X}_{cg}$.

Posterior and shrinkage. Conjugacy gives

$$\lambda_{cg} \mid Y_{cg} \sim \Gamma(\alpha_{cg} + Y_{cg}, \beta_{cg} + 1), \quad \hat{\lambda}_{cg}^{(\text{SX})} = \frac{Y_{cg} + \alpha_{cg}}{\beta_{cg} + 1}.$$

With $\beta_{cg} = 1/(\phi_g \tilde{X}_{cg})$ the posterior mean can be rewritten as

$$\hat{\lambda}_{cg}^{(\text{SX})} = w_{\text{obs}} Y_{cg} + w_{\text{pred}} \tilde{X}_{cg}, \quad w_{\text{obs}} = \frac{\phi_g \tilde{X}_{cg}}{1 + \phi_g \tilde{X}_{cg}}, \quad w_{\text{pred}} = 1 - w_{\text{obs}}.$$

Hence genes with large sample variance ($\phi_g \uparrow$) rely more on the noiseless AE prediction, echoing the empirical results in Section 3.

2.4 Posterior Inference and Imputation

Posterior predictive distribution. For each model $m \in \{\text{NI}, \text{Mix}, \text{SAV}, \text{SX}\}$ the latent expression λ_{cg} follows a Gamma posterior (cf. (3), (G2)). Integrating (L1) over this posterior yields an explicit *posterior predictive* pmf for the *next* observation Y_{cg}^* :

$$Y_{cg}^* \mid Y_{cg} = y \sim \begin{cases} \text{NB}(r = \alpha_{cg}^{(m)} + y, p = \frac{1}{\beta_{cg}^{(m)} + 1}), & m \in \{\text{NI}, \text{SAV}, \text{SX}\}, \\ \pi_{cg} \text{NB}(r^{(0)}, p^{(0)}) + (1 - \pi_{cg}) \text{NB}(r^{(1)}, p^{(1)}), & m = \text{Mix}, \end{cases}$$

where p is the success probability in the NB parameterisation and π_{cg} is the posterior responsibility from (E1). This closed form lets us sample without MCMC.

Imputation strategies. Two options are routinely used in the literature:

1. **Deterministic (posterior mean).** Replace the masked entry with $\hat{\lambda}_{cg}^{(m)}$ (Eqs. (4), (Est), (PM)), which minimises mean-squared error but underestimates variability.
2. **Stochastic (multiple imputation).** Draw M independent samples $y_{cg}^{*(1)}, \dots, y_{cg}^{*(M)}$ from the above NB / mixture-NB and set the imputed value to their average, or keep all M matrices to propagate uncertainty

Choice made in this work. To align with the evaluation protocol of Huang et al. [2], Wang et al. [3], we adopt strategy 1, thereby emulating “pseudo-observed counts” that retain realistic Poisson noise.

2.5 Evaluation Metrics

- a) **Coefficient of determination (R^2):** Measures how well a regression model predicts and explains the observed data. It represents the proportion of the variance in the dependent variable that’s predictable from the independent variable(s). Higher values (closer to 1.0) indicate a better fit.
- b) **Pearson correlation (r):** Measures the strength and direction of the linear relationship between two variables. Values range from -1 (perfect negative correlation) to +1 (perfect positive correlation), with 0 indicating no linear correlation. It assesses association, not causation.

- c) **Root Mean Square Error (RMSE)**: Measures the average magnitude of prediction errors in a model. It calculates the square root of the average of the squared differences between predicted values and actual values. Lower RMSE values indicate better model performance, as they reflect smaller prediction errors. It’s in the same units as the target variable.

Additionally, we visualise recovered versus ground-truth counts (scatter), Bland–Altman plots, and heatmaps of residuals for 500 randomly selected genes.

2.6 Software and Reproducibility

All analyses are implemented in `Python 3.12.6` using `NumPy`, `SciPy`, `scikit-learn`, and `TensorFlow 2.19.0` (for the autoencoder). Code and data processing scripts are available at Supplementary Materials.

3 Results

3.1 Quantitative Performance Across Dropout Rates

Table 1: Coefficient of determination (R^2) across dropout rates.

Dropout p	0.05	0.10	0.20	0.40	0.60	0.80
Non-informative	0.055055	0.109266	0.126136	0.041700	-0.114695	-0.349786
EM Mixture	-0.703851	-0.731692	-0.765339	-0.803190	-0.825137	-0.824153
SAVER	0.292095	-0.483019	-0.806474	-0.851395	-0.114695	-0.262693
SAVER-X	0.959215	0.927010	0.862648	0.695386	0.445156	0.115606

Table 2: Pearson correlation (r) across dropout rates.

Dropout p	0.05	0.10	0.20	0.40	0.60	0.80
Non-informative	0.527526	0.554633	0.563068	0.520850	0.442652	0.325107
EM Mixture	0.139534	0.124861	0.106429	0.081524	0.061468	0.040090
SAVER	0.646047	0.258490	0.096763	0.074302	0.442652	0.368653
SAVER-X	0.979607	0.963504	0.931323	0.84769	0.725345	0.557803

Table 3: Root Mean Square Error (RMSE) across dropout rates.

Dropout p	0.05	0.10	0.20	0.40	0.60	0.80
Non-informative	0.972055	0.943760	0.934780	0.978900	1.055761	1.161770
EM Mixture	1.305280	1.315901	1.328623	1.342791	1.350938	1.350574
SAVER	0.841347	1.217758	1.344013	1.360621	1.055761	1.123664
SAVER-X	0.201946	0.270159	0.370599	0.551903	0.737816	0.940395

3.2 Visual Diagnostics

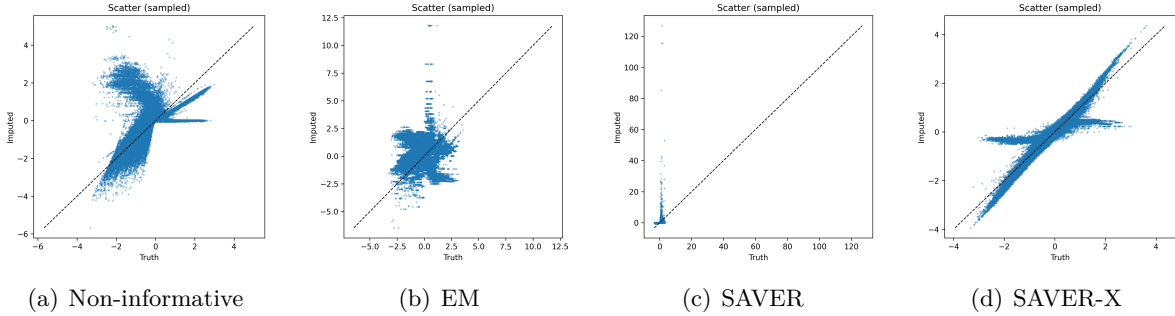


Figure 1: Scatter plot of recovered vs. true expression for $p = 0.20$

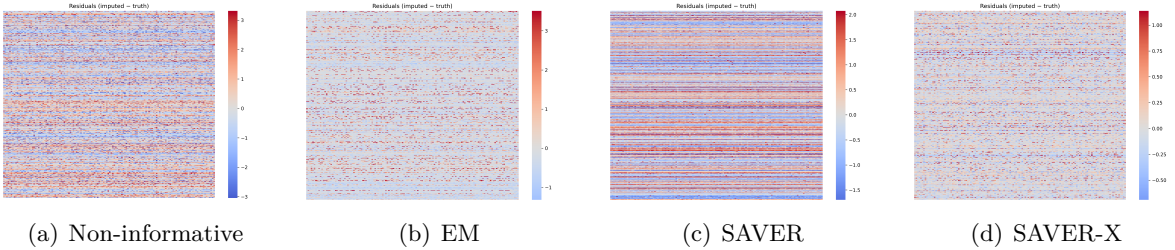


Figure 2: Heatmap of recovered vs. true expression for $p = 0.20$

Additional Bland–Altman and residual heatmaps are provided in Supplementary Materials.

4 Discussion

4.1 Interpretation of Findings

As seen in both quantitative and visual diagnostics, by adding additional information of healthy tissues, the autoencoder prior (SAVER-X) consistently outperforms other methods across dropout rates, particularly at higher dropout levels.

Interestingly, the non-informative prior shows competitive performance but is less robust than SAVER-X. At high dropout rates, it struggles with high dropout rates, leading to significant underestimation of true expression values.

The Poisson–Lasso prior (SAVER) performs well at moderate dropout rates at PCA plots, but its overall performance is hindered by its reliance on sparsity assumptions, which may not hold in bulk data. This leads to extreme outliers in the scatter plots and Bland–Altman Graphs, plus it shows low Pearson correlation and R^2 values. At high dropout rates, its performance does not degrade as much as the non-informative prior, but it still lags behind SAVER-X.

The two-component mixture prior (EM) performs the worst overall, especially at high dropout rates. It fails to show linearity in the scatter plots, indicating poor recovery of true expression values. At all dropout rates, it shows a significant bias towards zero, leading to low Pearson correlation and R^2 values. This suggests that the mixture model may not be suitable for bulk RNA-Seq data,

where the underlying distribution of expression values is more complex than a simple mixture of two components.

Overall, our results demonstrate that the superior performance of SAVER-X is largely attributable to its use of external information from healthy tissue data. In contrast, the SAVER model, which relies on a Poisson–Lasso prior, struggles to handle outliers effectively; its sparsity assumption leads to extreme deviations in the imputed values, as reflected by the presence of outliers in diagnostic plots and reduced correlation metrics. The EM-based two-component mixture prior, while effective in single-cell RNA-Seq where cell-to-cell heterogeneity is prominent, appears ill-suited for bulk RNA-Seq. These findings highlight the importance of leveraging informative priors and tailoring model assumptions to the specific characteristics of bulk RNA-Seq data.

4.2 Limitations

- Limited posterior sampling may underestimate variance, leading to accidental overconfidence in model predictions.
- Only one bulk dataset analysed; generalisability remains untested.
- Training data available solely to SAVER-X; future studies should ensure fair competition, introducing additional datasets for the other three models.

4.3 Future Work

- Incorporate multiple posterior draws and Bayesian model averaging.
- Explore additional evaluation metrics which uniformly assess model performance.
- Explore hierarchical priors that share information across genes, especially for EM and SAVER.
- Evaluate on additional cancer and non-cancer bulk datasets.
- Extend framework to account for batch effects and covariates.

5 Conclusion

In this study, we systematically evaluated four Bayesian priors for bulk RNA-Seq expression recovery within a unified Poisson–Gamma framework. Our findings indicate that the autoencoder prior (SAVER-X) consistently outperforms other methods by leveraging external information from healthy tissue data. The non-informative prior shows competitive performance but is less robust, while the Poisson–Lasso prior (SAVER) struggles with outliers and sparsity assumptions. The two-component mixture prior (EM) performs the worst, indicating its unsuitability for bulk RNA-Seq data.

These results highlight the importance of prior selection and model assumptions in bulk RNA-Seq imputation, providing insights into best practices for future studies. Our work contributes to the understanding of Bayesian methods in bulk RNA-Seq and sets the stage for further research in this area.

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